

7

The Respiratory System and Its Regulation

In this chapter and in the web study guide

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Inspiration

Expiration

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ACTIVITY 7.4 *Regulation of Pulmonary Ventilation* provides an in-depth review of the involuntary regulation of pulmonary ventilation.

In Closing

By any standard, Beijing, China, is one of the most polluted cities on the planet. In preparation for the 2008 Olympic Games, nearly \$17 billion was spent in attempts to temporarily improve air quality, including cloud seeding to increase the likelihood of rain showers in the region overnight. Factories were closed, traffic was halted, and construction was put on hold for the duration of the Games. Yet air pollution at the Olympics was still about two to four times higher than that of Los Angeles on an average day, exceeding levels considered safe by the World Health Organization. Several athletes opted out of events because of respiratory problems or concerns, including Ethiopian marathon record holder Haile Gebrselassie and 2004 cycling silver medalist Sérgio Paulinho of Portugal. Athletes previously diagnosed with asthma were allowed to use rescue inhalers. For the first time ever, soccer matches were interrupted to give athletes time to recover from the pollutants, smog, heat, and humidity. Athletes and spectators endured these conditions for a few weeks, and there are no reports of long-term health problems among athletes or spectators from exposure to the Beijing air. However, the residents of Beijing encounter these adverse respiratory conditions on a daily basis.

The respiratory and cardiovascular systems combine to provide an effective delivery system that carries oxygen to, and removes carbon dioxide from, all tissues of the body.

This transportation involves four separate processes:

- Pulmonary ventilation (breathing): movement of air into and out of the lungs
- Pulmonary diffusion: the exchange of oxygen and carbon dioxide between the lungs and the blood
- Transport of oxygen and carbon dioxide via the blood
- Capillary diffusion: the exchange of oxygen and carbon dioxide between the capillary blood and metabolically active tissues

The first two processes are referred to as **external respiration** because they involve moving gases from outside the body into the lungs and then the blood. Once the gases are in the blood, they must be transported to the tissues. When blood arrives at the tissues, the fourth step of respiration occurs. This gas exchange between the

blood and the tissues is called **internal respiration**. Thus, external and internal respiration are linked by the circulatory system. The following sections examine all four components of respiration.

Pulmonary Ventilation

Pulmonary ventilation, or breathing, is the process by which we move air into and out of the lungs. The anatomy of the respiratory system is illustrated in [figure 7.1](#). At rest, air is typically drawn into the lungs through the nose, although the mouth must also be used when the demand for air exceeds the amount that can comfortably be brought in through the nose. Nasal breathing is advantageous because the air is warmed and humidified as it swirls through the bony irregular sinus surfaces (turbinates or conchae). Of equal importance, the turbinates churn the inhaled air, causing dust and other particles to contact and adhere to the nasal mucosa. This filters out all but the tiniest particles, minimizing irritation and the threat of respiratory infections. From the nose and mouth, the air travels through the pharynx, larynx, trachea, and bronchial tree.

This transport zone also has physiological significance because it comprises the so-called anatomical **dead space**. Because part of each expired breath stays within this space, air from outside the body mixes with this air with each inspiration, and the resulting mixture reaches the alveoli.

These anatomical structures serve a transport function only, because gas exchange does not occur in these structures. Exchange of oxygen and carbon dioxide occurs when air finally reaches the smallest respiratory units: the respiratory bronchioles and the alveoli. The respiratory bronchioles are primarily transport tubes also but are included in this region because they contain clusters of alveoli. This is known as the respiratory zone because it is the site of gas exchange in the lungs.

The lungs are not directly attached to the ribs. Rather, they are suspended by the pleural sacs. The pleural sacs have a double wall: the parietal pleura, which lines the thoracic wall, and the visceral or pulmonary pleura, which lines the outer aspects of the lung. These pleural walls envelop the lungs and have a thin film of fluid between them that reduces friction during respiratory movements. In addition,

these sacs are connected to the lungs and the inner surface of the thoracic cage, causing the lungs to take the shape and size of the rib or thoracic cage as the chest expands and contracts.

The anatomy of the lungs, the pleural sacs, the diaphragm muscle, and the thoracic cage determines airflow into and out of the lungs, that is, inspiration and expiration.

Inspiration

Inspiration is an active process involving the diaphragm and the external intercostal muscles. [Figure 7.2a](#) shows the resting positions of the diaphragm and the thoracic cage, or thorax. With inspiration, the ribs and sternum are moved by the external intercostal muscles. The ribs swing up and out and the sternum swings up and forward. At the same time, the diaphragm contracts, flattening down toward the abdomen.

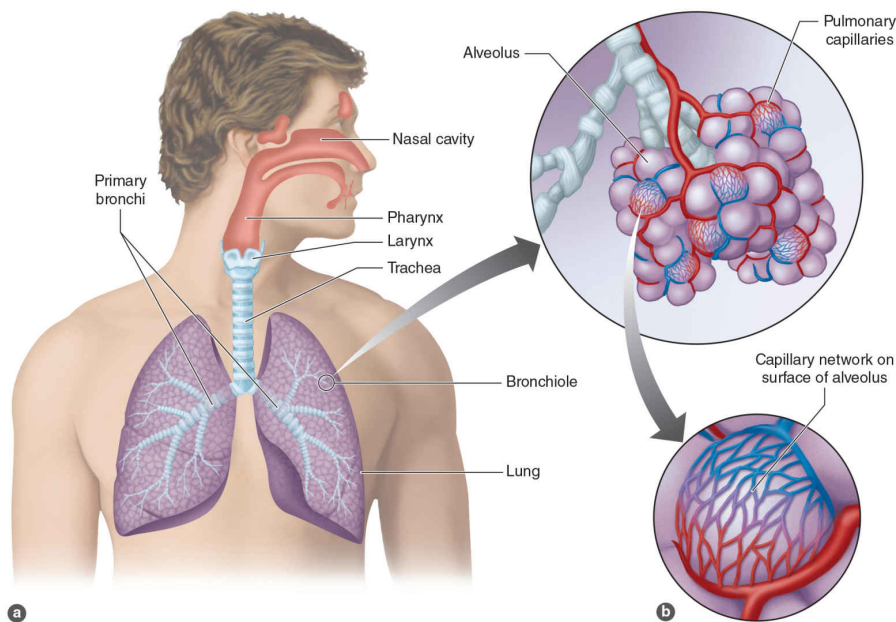


FIGURE 7.1 (a) The anatomy of the respiratory system, illustrating the respiratory tract (i.e., nasal cavity, pharynx, trachea, and bronchi). (b) The enlarged view of an alveolus shows the regions of gas exchange between the alveolus and pulmonary blood in the capillaries.

These actions, illustrated in [figure 7.2b](#), expand all three dimensions of the thoracic cage, increasing the volume inside the lungs. When the lungs are expanded they have a greater volume, and the air within them has more space to fill. According to **Boyle's**

gas law, which states that pressure $\times$ volume is constant (at a constant temperature), the pressure within the lungs decreases. As a result, the pressure in the lungs (intrapulmonary pressure) is less than the air pressure outside the body. Because the respiratory tract is open to the outside, air rushes into the lungs to reduce this pressure difference. This is how air moves into the lungs during inspiration.

The pressure changes required for adequate ventilation at rest are really quite small. For example, at the standard atmospheric pressure at sea level (760 mmHg), inspiration may decrease the pressure in the lungs (intrapulmonary pressure) by only about 2 to 3 mmHg. However, during maximal respiratory effort, such as during exhaustive exercise, the intrapulmonary pressure can decrease by 80 to 100 mmHg.

During forced or labored breathing, as during heavy exercise, inspiration is further assisted by the action of other muscles, such as the scalenes (anterior, middle, and posterior) and sternocleidomastoid in the neck and the pectorals in the chest. These muscles help raise the ribs even more than during regular breathing.

Expiration

At rest, **expiration** is a passive process involving relaxation of the inspiratory muscles and elastic recoil of the lung tissue. As the diaphragm relaxes, it returns to its normal upward, arched position. As the external intercostal muscles relax, the ribs and sternum move back into their resting positions ([figure 7.2c](#)). While this happens, the elastic nature of the lung tissue causes it to recoil to its resting size. This increases the pressure in the lungs and causes a proportional decrease in volume in the thorax, and therefore air is forced out of the lungs.

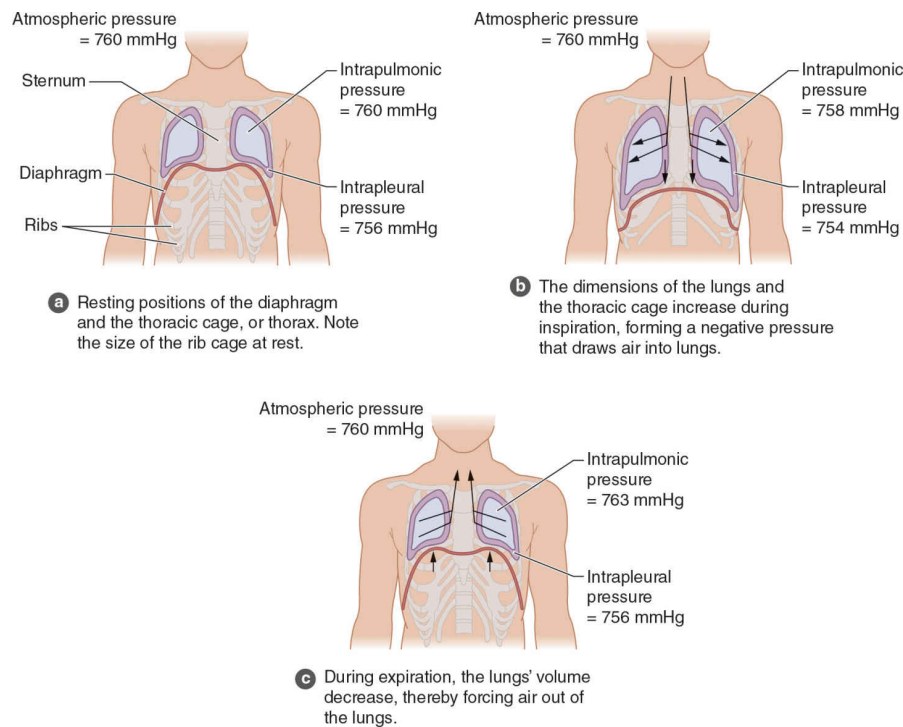


FIGURE 7.2 The process of inspiration and expiration, showing (a) the positions of the ribs and thorax at rest, and how movement of the ribs and diaphragm (b) increase the size of the thorax during inspiration and (c) decrease the size of the thorax during expiration.

During forced breathing, expiration becomes a more active process. The internal intercostal muscles actively pull the ribs down. This action can be assisted by the latissimus dorsi and quadratus lumborum muscles. Contracting the abdominal muscles increases the intra-abdominal pressure, forcing the abdominal viscera upward against the diaphragm and accelerating its return to the domed position. These muscles also pull the rib cage down and inward.

The changes in intra-abdominal and intrathoracic pressure that accompany forced breathing also help return venous blood back to the heart, working together with the muscle pump in the legs to assist the return of venous volume. As intra-abdominal and intrathoracic pressure increases, it is transmitted to the great veins—the pulmonary veins and superior and inferior venae cavae—that transport blood back to the heart. When the pressure decreases, the veins return to their original size and fill with blood. The changing pressures within the abdomen and thorax squeeze the blood in the veins, assisting its return through a milking action. This phenomenon

is known as the **respiratory pump** and is essential in maintaining adequate venous return.

Pulmonary Volumes

The volume of air in the lungs can be measured with a technique called **spirometry**. A spirometer measures the volumes of air inspired and expired and therefore changes in lung volume. Although more sophisticated spirometers are used today, a simple spirometer contains a bell filled with air that is partially submerged in water. A tube runs from the subject's mouth under the water and emerges inside the bell, just above the water level. As the person exhales, air flows down the tube and into the bell, causing the bell to rise. The bell is attached to a pen, and its movement is recorded on a simple rotating drum ([figure 7.3](#)).

This technique is used clinically to measure lung volumes, capacities, and flow rates as an aid in diagnosing such respiratory diseases as asthma, chronic obstructive pulmonary disease (COPD), and emphysema.

The amount of air entering and leaving the lungs with each breath is known as the **tidal volume**. The **vital capacity (VC)** is the greatest amount of air that can be expired after a maximal inspiration. Even after a full expiration, some air remains in the lungs. The amount of air remaining in the lungs after a maximal expiration is the **residual volume (RV)**. The RV cannot be measured with spirometry. The **total lung capacity (TLC)** is the sum of the VC and the RV.

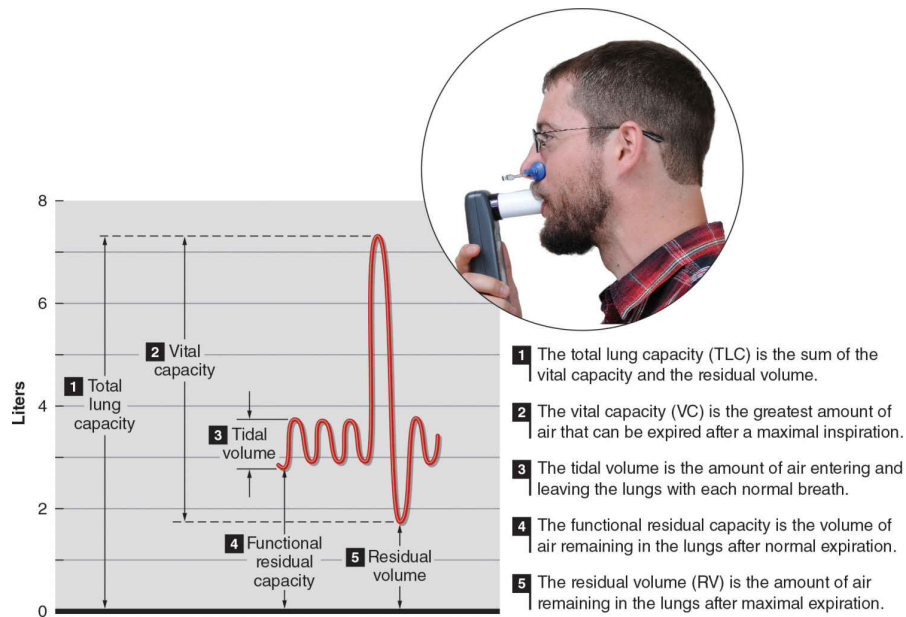


FIGURE 7.3 Lung volumes measured by spirometry.

In Review

- Pulmonary ventilation (breathing) is the process by which air is moved into and out of the lungs. It has two phases: inspiration and expiration.
- Inspiration is an active process in which the diaphragm and the external intercostal muscles contract, increasing the dimensions, and thus the volume, of the thoracic cage. This decreases the pressure in the lungs, causing air to flow in.
- Expiration at rest is normally a passive process. The inspiratory muscles and diaphragm relax and the elastic tissue of the lungs recoils, returning the thoracic cage to its smaller, normal dimensions. This increases the pressure in the lungs and forces air out.
- The pressure changes required for ventilation at rest are small, as little as 2 mmHg. However, during maximal respiratory effort, the intrapulmonary pressure can decrease by 80 to 100 mmHg.
- Forced or labored inspiration and expiration are active processes and involve accessory muscle actions.
- Breathing through the nose helps humidify and warm the air during inhalation and filters out foreign particles from the air. Mouth breathing dominates at moderate to high exercise intensities.
- Lung volumes and capacities, along with rates of airflow into and out of the lungs, are measured by spirometry.

Pulmonary Diffusion

Gas exchange in the lungs between the alveoli and the capillary blood, called **pulmonary diffusion**, serves two major functions:

- It replenishes the blood's oxygen supply, which is depleted at the tissue level as it is used for oxidative energy production.
- It removes carbon dioxide from venous blood returning from systemic tissues.

Air is brought into the lungs during pulmonary ventilation, enabling gas exchange to occur through pulmonary diffusion. Oxygen from the air diffuses from the alveoli into the blood in the pulmonary capillaries, and carbon dioxide diffuses from the blood into the alveoli in the lungs. The **alveoli** are grapelike clusters, or air sacs, at the ends of the terminal bronchioles.

RESEARCH PERSPECTIVE 7.1

Sprint Interval Training for Respiratory Muscles

Typical respiratory muscle endurance training (RMET) improves exercise capacity and thus performance; such improvements are largely attributed to reductions in the development of respiratory muscle fatigue. However, can a shorter version of RMET based on the principle of high-intensity interval training (respiratory muscle sprint-interval training, or RMSIT) elicit similar improvements in respiratory muscle function?

A team of investigators recently sought to compare the effects of traditional RMET versus RMSIT on respiratory muscle function.⁷ Mechanical airway properties and respiratory muscle testing (e.g., respiratory muscle strength) were measured before and after experimental sessions of RMET and RMSIT. RMET consisted of continuous volitional hyperpnea (increased depth and rate of breathing) performed for 30 min using a commercially available training device. The RMSIT was a novel respiratory muscle training regimen developed by the researchers. Using the same training device as with RMET, the RMSIT regimen consisted of six short maximal respiratory sprints with additional airway resistance to maximize respiratory muscle work. In this fashion, RMSIT is characterized by higher respiratory muscle power output and tension-time indices, but considerably lower total work compared to RMET. The standard RMET and the novel RMSIT regimens reduced respiratory muscle contractility to the same extent, triggering similar muscular adaptations in response to training. Neither protocol altered mechanical airway properties. Therefore, RMSIT appears to be a safe and time-saving alternative to RMET.

RMET can improve overall function for individuals who have undergone a median sternotomy (splitting of the sternum to access underlying organs) during cardiac surgery. Clinical exercise physiologists are interested in exercise training adaptations that occur with structured cardiac rehabilitation programs. The results of a 2013 study suggest that it would be beneficial to include exercises that improve the strength of the inspiratory muscles as part of a cardiac rehabilitation program.⁴ This type of training would reduce inspiratory muscle effort and further improve ventilatory efficiency in patients after open-heart surgery.

Blood from the body (except for that returning from the lungs) returns through the vena cava to the right side of the heart. From the right ventricle, this blood is pumped through the pulmonary artery to the lungs, ultimately working its way into the pulmonary capillaries. These capillaries form a dense network around the alveolar sacs and are so small that the red blood cells must pass through them in single file, such that the maximal surface area of each cell is exposed to the surrounding lung tissue. This is where pulmonary diffusion occurs.

Blood Flow to the Lungs at Rest

At rest the lungs receive approximately 4 to 6 L/min of blood flow, depending on body size. Because cardiac output from the right side of the heart approximates cardiac output from the left side of the heart, blood flow to the lungs matches blood flow to the systemic circulation. However, pressure and vascular resistance in the blood vessels in the lungs are different than in the system circulation. The mean pressure in the pulmonary artery is ~15 mmHg (systolic pressure is ~25 mmHg and diastolic pressure is ~8 mmHg) compared to the mean pressure in the aorta of ~95 mmHg. The pressure in the left atrium, where blood is returning to the heart from the lungs, is ~5 mmHg; thus, there is not a great pressure difference across the pulmonary circulation (15 – 5 mmHg). [Figure 7.4](#) illustrates the differences in pressures between the pulmonary and systemic circulation.

Recalling the discussion of blood flow in the cardiovascular system from [chapter 6](#), $\text{pressure} = \text{flow} \times \text{resistance}$. Since blood flow to the lungs is equal to that of the systemic circulation, and there is a substantially lower change in pressure across the pulmonary vascular system, resistance is proportionally lower compared to that in the

systemic circulation. This is reflected in differences in the anatomy of the vessels in the pulmonary versus systemic circulation: The pulmonary blood vessels are thin walled, with relatively little smooth muscle.

Respiratory Membrane

Gas exchange between the air in the alveoli and the blood in the pulmonary capillaries occurs across the **respiratory membrane** (also called the alveolar-capillary membrane). This membrane, depicted in [figure 7.5](#), is composed of

- the alveolar wall,
- the capillary wall, and
- their respective basement membranes.

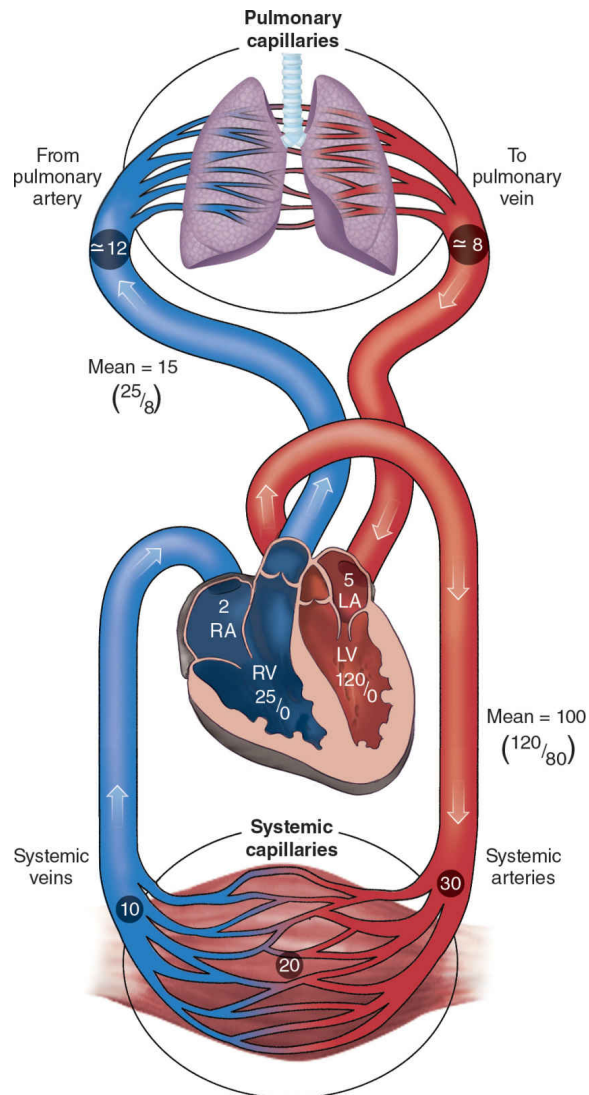


FIGURE 7.4 Comparison of pressures (mmHg) in the pulmonary and systemic circulations.



The primary function of these membranous surfaces is for gas exchange. The respiratory membrane is very thin, measuring only 0.5 to 4 mm. As a result, the gases in the nearly 300 million alveoli are in close proximity to the blood circulating through the capillaries.

Partial Pressures of Gases

The air we breathe is a mixture of gases. Each exerts a pressure in proportion to its concentration in the gas mixture. The individual pressures from each gas in a mixture are referred to as **partial pressures**. According to **Dalton's law**, the total pressure of a mixture of gases equals the sum of the partial pressures of the individual gases in that mixture.

Consider the air we breathe. It is composed of 79.04% nitrogen (N_2), 20.93% oxygen (O_2), and 0.03% carbon dioxide (CO_2). These percentages remain constant regardless of altitude. At sea level, the atmospheric (or barometric) pressure is approximately 760 mmHg, which is also referred to as standard atmospheric pressure. Thus, if the total atmospheric pressure is 760 mmHg, then the partial pressure of nitrogen (P_{N_2}) in air is 600.7 mmHg (79.04% of the total 760 mmHg pressure). Oxygen's partial pressure (P_{O_2}) is 159.1 mmHg (20.93% of 760 mmHg), and carbon dioxide's partial pressure (P_{CO_2}) is 0.2 mmHg (0.03% of 760 mmHg).

In the human body, gases are usually dissolved in fluids, such as blood plasma. According to **Henry's law**, gases dissolve in liquids in proportion to their partial pressures, depending also on their solubilities in the specific fluids and on the temperature. A gas's solubility in blood is a constant, and blood temperature also remains relatively constant at rest. Thus, the most critical factor for gas exchange between the alveoli and the blood is the pressure gradient between the gases in the two areas.

Gas Exchange in the Alveoli

Differences in the partial pressures of the gases in the alveoli and the gases in the blood create a pressure gradient across the respiratory membrane. This forms the basis of gas exchange during pulmonary diffusion. If the pressures on each side of the membrane were equal, the gases would be at equilibrium and would not move. But the pressures are not equal, so gases move according to partial pressure gradients.

Oxygen Exchange

The PO_2 of air outside the body at standard atmospheric pressure is 159 mmHg. But this pressure decreases to about 105 mmHg when air is inhaled and enters the alveoli, where it is moistened and mixes

with the air in the alveoli. The alveolar air is saturated with water vapor (which has its own partial pressure) and contains more carbon dioxide than the inspired air. Both the increased water vapor pressure and increased partial pressure of carbon dioxide contribute to the total pressure in the alveoli. Fresh air that ventilates the lungs is constantly mixed with the air in the alveoli while some of the alveolar gases are exhaled to the environment. As a result, alveolar gas concentrations remain relatively stable.

The blood, stripped of much of its oxygen by the metabolic needs of the tissues, typically enters the pulmonary capillaries with a PO_2 of about 40 mmHg (see [figure 7.6](#)). This is about 60 to 65 mmHg less than the PO_2 in the alveoli. In other words, the pressure gradient for oxygen across the respiratory membrane is typically about 65 mmHg. As noted earlier, this pressure gradient drives the oxygen from the alveoli into the blood to equilibrate the pressure of the oxygen on each side of the membrane.

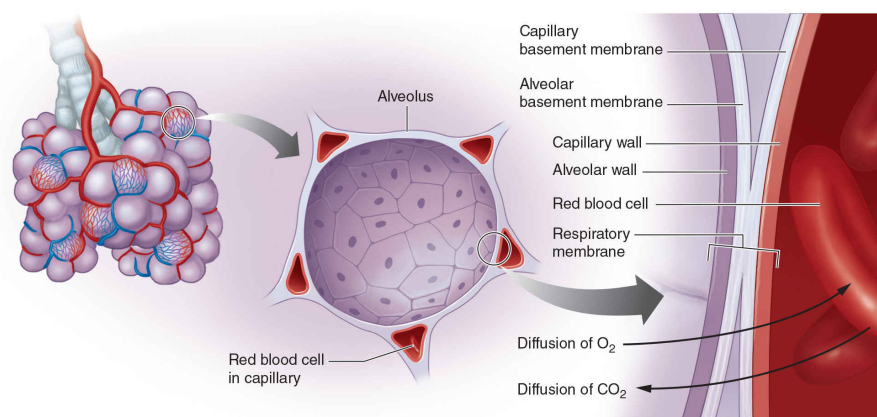


FIGURE 7.5 The anatomy of the respiratory membrane, showing the exchange of oxygen and carbon dioxide between an alveolus and pulmonary capillary blood.

RESEARCH PERSPECTIVE 7.2

Exercise Training Offsets Decreases in Lung Diffusing Capacity with Aging

The structure and function of the pulmonary vasculature contributes to maximal aerobic capacity ($\dot{V}O_{2max}$), such that a larger, more distensible vascular network in the lungs is associated with greater aerobic exercise capacity. During exercise, increased cardiac output and pulmonary perfusion pressure cause an expansion of the highly compliant pulmonary capillary

network, resulting in increased lung diffusing capacity, alveolar-capillary membrane conductance, and pulmonary capillary blood volume.

As we age, the structure and function of the pulmonary circulation changes, resulting in increased pulmonary vascular stiffness, pulmonary vascular pressures, and pulmonary vascular resistance, all of which impair recruitment and distension of pulmonary capillaries during exercise. However, these age-related alterations do not appear to limit the expansion of pulmonary capillaries during exercise in healthy older adults. The pulmonary vascular response to exercise in endurance-trained, highly fit older adults is not well defined. It is plausible that a higher $\dot{V}O_{2max}$ may cause the demand for cardiac output and pulmonary blood flow during exercise to remain elevated in older athletes, thus predisposing highly fit older adults to impairments in pulmonary vascular expansion and pulmonary gas exchange relative to the metabolic demands of exercise.

This concept was recently tested by a group of investigators who characterized lung diffusing capacity, alveolar-capillary membrane conductance, and pulmonary capillary blood volume in response to incremental exhaustive exercise in aerobically trained older adults.³ The authors hypothesized that older athletes would be limited in their ability to expand the pulmonary vascular network during high-intensity exercise. Their findings confirmed the negative age-related reductions in lung diffusing capacity, alveolar-capillary membrane conductance, and pulmonary capillary blood volume during exercise; however, these variables were increased in exercise-trained older adults during exercise relative to age-matched, nontrained individuals. In contrast to the original hypothesis, there was a progressive increase in lung diffusing capacity throughout exercise in exercise-trained adults, suggesting that the expansion of the pulmonary capillary network during exercise is not limited during exercise in highly fit older adults. Follow-up studies should include measures of pulmonary vascular pressures to more specifically determine the relation between increases in lung diffusing capacity and the pulmonary vascular response to exercise.

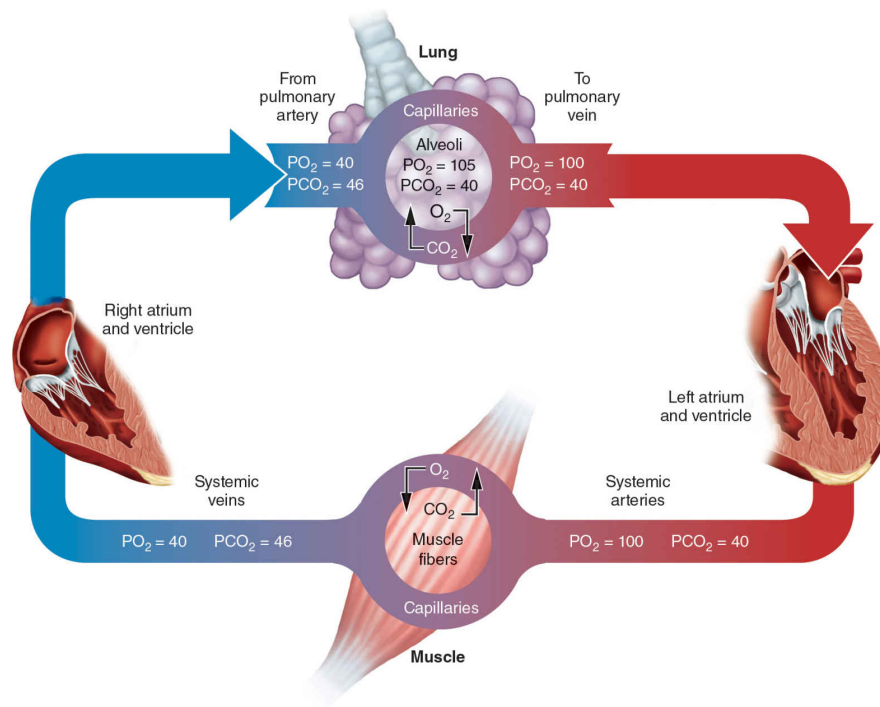


FIGURE 7.6 Partial pressure of oxygen (PO_2) and carbon dioxide (PCO_2) in blood as a result of gas exchange in the lungs and gas exchange between the capillary blood and tissues.



The PO_2 in the alveoli stays relatively constant at about 105 mmHg. As the deoxygenated blood enters the pulmonary artery, the PO_2 in the blood is only about 40 mmHg. But as the blood moves along the pulmonary capillaries, gas exchange occurs. By the time the pulmonary blood reaches the venous end of these capillaries, the PO_2 in the blood equals that in the alveoli (approximately 105 mmHg), and the blood is now considered to be saturated with oxygen at its full carrying capacity. The blood leaving the lungs through the pulmonary veins and subsequently returning to the systemic (left) side of the heart has a rich supply of oxygen to deliver to the tissues. Notice,

however, that the PO_2 in the pulmonary vein is 100 mmHg, not the 105 mmHg found in the alveolar air and pulmonary capillaries. This difference is attributable to the fact that about 2% of the blood is shunted from the aorta directly to the lung to meet the oxygen needs of the lung itself. This blood has a lower PO_2 and reenters the pulmonary vein along with fully saturated blood returning to the left atrium that has just completed gas exchange. This blood mixes and thus decreases the PO_2 of the blood returning to the heart.

Diffusion through tissues is described by **Fick's law** (figure 7.7). Fick's law states that the rate of diffusion through a tissue such as the respiratory membrane is proportional to the surface area and the difference in the partial pressure of gas between the two sides of the tissue. For example, the greater the pressure gradient for oxygen is across the respiratory membrane, the more rapidly oxygen diffuses across it. The rate of diffusion is also inversely proportional to the thickness of the tissue in which the gas must diffuse. Additionally, the diffusion constant, which is unique to each gas, influences the rate of diffusion across the tissue. Carbon dioxide has a much lower diffusion constant than oxygen; therefore, even though there is not as great a difference between alveolar and capillary partial pressure of carbon dioxide as there is for oxygen, carbon dioxide still diffuses easily.

The rate at which oxygen diffuses from the alveoli into the blood is referred to as the **oxygen diffusion capacity** and is expressed as the volume of oxygen that diffuses through the membrane each minute for a pressure difference of 1 mmHg. At rest, the oxygen diffusion capacity is about 21 ml of oxygen per minute per 1 mmHg of pressure difference between the alveoli and the pulmonary capillary blood. Although the partial pressure gradient between venous blood coming into the lung and the alveolar air is about 65 mmHg (105 mmHg – 40 mmHg), the oxygen diffusion capacity is calculated on the basis of the mean pressure in the pulmonary capillary, which has a substantially higher PO_2 . The gradient between the mean partial pressure of the pulmonary capillary and the alveolar air is approximately 11 mmHg, which would provide a diffusion of 231 ml of oxygen per minute through the respiratory membrane. During maximal exercise, the oxygen diffusion capacity may increase by up to three times the resting rate, because blood is returning to the lungs

severely desaturated and thus there is a greater partial pressure gradient from the alveoli to the blood. In fact, rates of more than 80 ml/min have been observed among highly trained athletes.

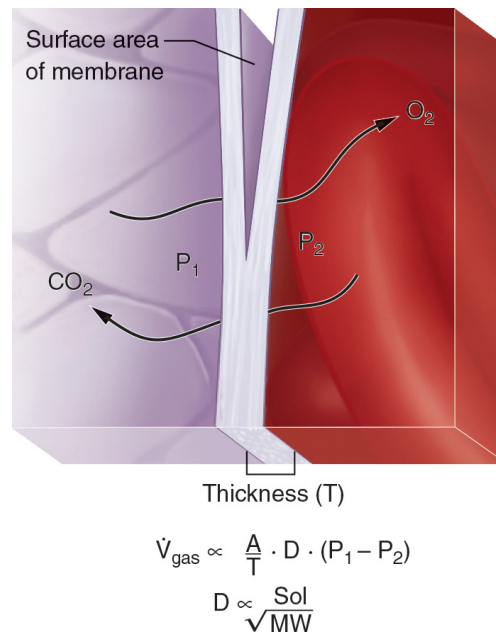
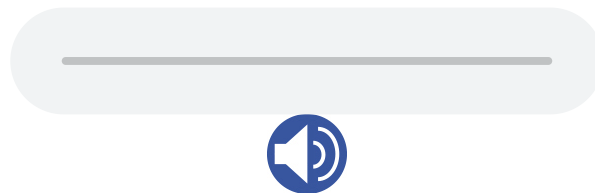


FIGURE 7.7 Diffusion through a sheet of tissue. The amount of gas ($\dot{V}_{\text{gas}}$) transferred is proportional to the area (A), a diffusion constant (D), and the difference in partial pressure ($P_1 - P_2$) and is inversely proportional to the thickness (T). The constant is proportional to the gas solubility (Sol) but inversely proportional to the square root of its molecular weight (MW).



The increase in oxygen diffusion capacity from rest to exercise is caused by a relatively inefficient, sluggish circulation through the lungs at rest, which results primarily from limited perfusion of the upper regions of the lungs attributable to gravity. If the lung is divided into three zones as depicted in [figure 7.8](#), at rest only the bottom third (zone 3) of the lung is perfused with blood. During exercise, however, blood flow through the lungs is greater, primarily as a result of elevated blood pressure, which increases lung perfusion.

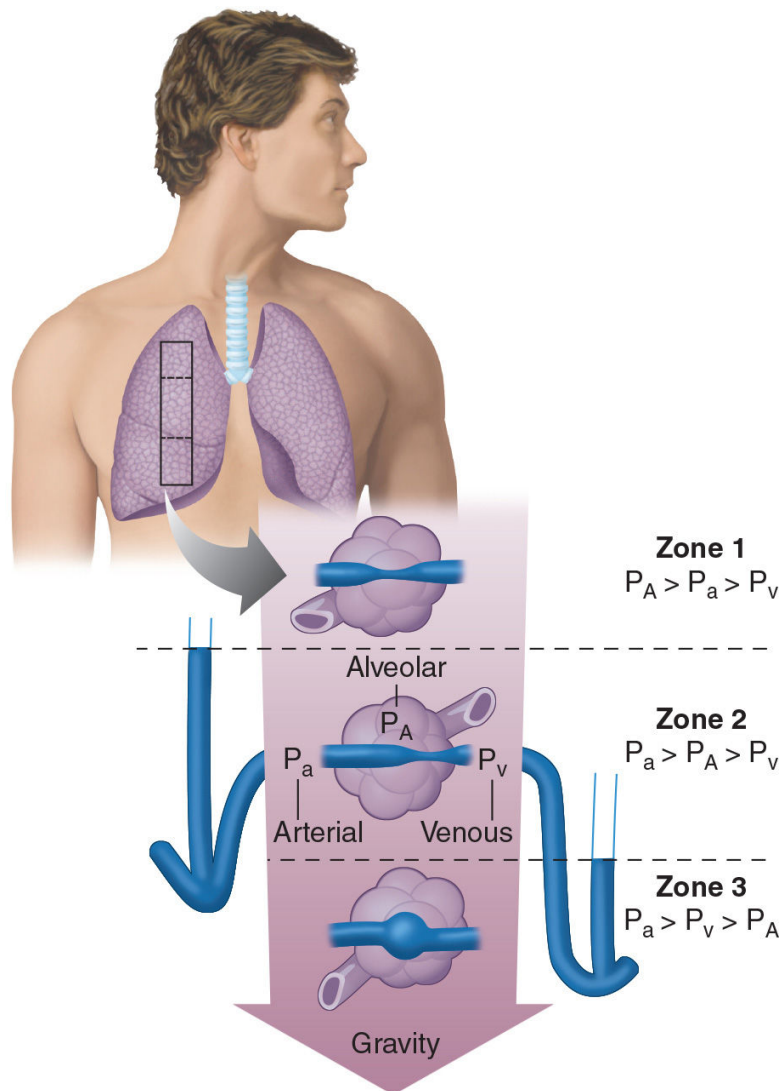


FIGURE 7.8 Explanation of the uneven distribution of blood flow in the lung.

Carbon Dioxide Exchange

Carbon dioxide, like oxygen, moves along a partial pressure gradient. As shown in [figure 7.6](#), the blood passing from the right side of the heart through the alveoli has a PCO_2 of about 46 mmHg. Air in the alveoli has a PCO_2 of about 40 mmHg. Although this results in a relatively small pressure gradient of only about 6 mmHg, it is more than adequate to allow for exchange of CO_2 . Carbon dioxide's diffusion coefficient is 20 times greater than that of oxygen, so CO_2 can diffuse across the respiratory membrane much more rapidly.

Summary of Pulmonary Gas Diffusion

The partial pressures of gases involved in pulmonary diffusion are summarized in [table 7.1](#). Note that the total pressure in the venous blood is only 706 mmHg, 54 mmHg lower than the total pressure in dry air and alveolar air. This is the result of a much greater decrease in PO_2 compared with the increase in PCO_2 as the blood goes through the body's tissues.

TABLE 7.1 Partial Pressures of Respiratory Gases at Sea Level

Gas	Partial pressure (mmHg)					
	% in dry air	Dry air	Alveolar air	Arterial blood	Venous blood	Diffusion gradient
H ₂ O	0	0	47	47	47	0
O ₂	20.93	159.1	105	100	40	60
CO ₂	0.03	0.2	40	40	46	6
N ₂	79.04	600.7	568	573	573	0
Total	100.00	760	760	760	706 ^a	0

^aSee text for an explanation of the decrease in total pressure.

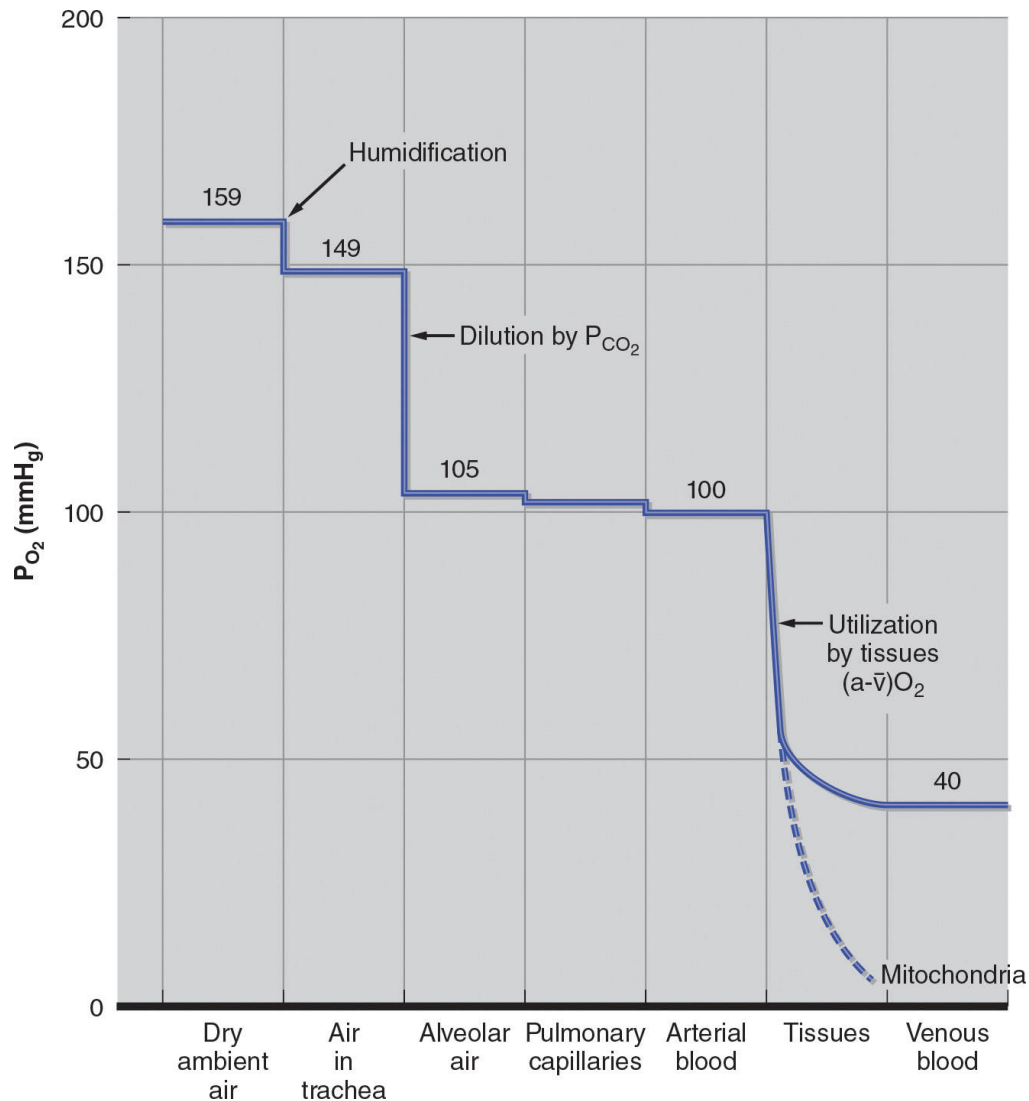


FIGURE 7.9 The oxygen cascade depicts the dropping partial pressures of oxygen (in this depiction, at sea level) from dry ambient air to the tissues and into the venous circulation draining those tissues.



Figure 7.9 shows the dropping partial pressures of oxygen at sea level from dry ambient air to the tissues and into the venous circulation draining those tissues. This is referred to as the **oxygen cascade**. At a sea level barometric pressure (P_B) of 760 mmHg, PO_2

in the ambient air (if it were completely devoid of moisture, which does not occur in nature) would be

$$0.2093 \times 760 \text{ mmHg} = 159 \text{ mmHg}.$$

As dry air moves through the nose and mouth and becomes humidified water vapor (which has a partial pressure, P_{H_2O} , of 47 mmHg at body temperature), air in the trachea has a partial pressure of

$$0.2093 \times (760 - 47) = 149 \text{ mmHg}.$$

In the alveoli, air now becomes a mixture combining PCO_2 in blood returning from the systemic circulation and PO_2 from the tracheal air and equilibrates at approximately 105 mmHg. As oxygen diffuses from the alveoli into the pulmonary capillaries and into arterial blood, PO_2 continues to drop slightly down diffusion gradients, since pulmonary capillary blood is a mixture of arterial and venous blood, a so-called admixture.

At the tissue (e.g., muscle) level, cells extract O_2 from the arterial supply for aerobic metabolism, and the drop in PO_2 from arterial blood to venous blood flowing away from the tissues represents the arterial–venous oxygen difference, or (a-v) O_2 difference. Note that the PO_2 at the mitochondrial level is extremely low, approximately 1 to 2 mmHg. This ensures optimal O_2 delivery to these organelles, the ultimate destination of oxygen where it is used in oxidative phosphorylation.

In Review

- Pulmonary diffusion is the process by which gases are exchanged across the respiratory membrane in the alveoli.
- Dalton's law states that the total pressure of a mixture of gases equals the sum of the partial pressures of the individual gases in that mixture.
- The amount and rate of gas exchange that occur across the membrane depend primarily on the partial pressure of each gas, although other factors are also important, as shown by Fick's law. Gases diffuse along a pressure gradient, moving from an area of higher pressure to one of lower pressure. Thus, oxygen enters the blood and carbon dioxide leaves it.

- Oxygen diffusion capacity increases as one moves from rest to exercise. When exercising muscles require more oxygen to be used in the metabolic processes, venous oxygen is depleted and oxygen exchange at the alveoli is facilitated.
 - The pressure gradient for carbon dioxide exchange is less than for oxygen exchange, but carbon dioxide's diffusion coefficient is 20 times greater than that of oxygen, so carbon dioxide crosses the membrane readily without a large pressure gradient.
-

Transport of Oxygen and Carbon Dioxide in the Blood

We have considered how air moves into and out of the lungs via pulmonary ventilation and how gas exchange occurs via pulmonary diffusion. Next we consider how gases are transported in the blood to deliver oxygen to the tissues and remove the carbon dioxide that the tissues produce.

Oxygen Transport

Oxygen is transported by the blood either (1) combined with hemoglobin in the red blood cells (greater than 98%) or (2) dissolved in the blood plasma (less than 2%). Only about 3 ml of oxygen is dissolved in each liter of plasma. Assuming a total plasma volume of 3 to 5 L, only about 9 to 15 ml of oxygen can be carried in the dissolved state. This limited amount of oxygen cannot adequately meet the needs of even resting body tissues, which generally require more than 250 ml of oxygen per minute (depending on body size). However, hemoglobin, a protein contained within each of the body's 4 to 6 billion red blood cells, allows the blood to transport nearly 70 times more oxygen than can be dissolved in plasma.

Hemoglobin Saturation

As just noted, over 98% of oxygen is transported in the blood bound to hemoglobin. Each molecule of hemoglobin can carry four molecules of oxygen. When oxygen binds to hemoglobin, it forms oxyhemoglobin; hemoglobin that is not bound to oxygen is referred to as deoxyhemoglobin. The binding of oxygen to hemoglobin depends on the PO_2 in the blood and the bonding strength, or affinity, between hemoglobin and oxygen. The curve in [figure 7.10](#) is an oxygen–

hemoglobin dissociation curve, which shows the amount of hemoglobin saturated with oxygen at different PO_2 values. The shape of the curve is extremely important for its function in the body. The relatively flat upper portion means that at high PO_2 concentrations, such as in the lungs, large drops in PO_2 result in only small changes in hemoglobin saturation. This is called the “loading” portion of the curve. A high blood PO_2 results in almost complete hemoglobin saturation, which means that the maximal amount of oxygen is bound. But as the PO_2 decreases, so does hemoglobin saturation.

The steep portion of the curve coincides with PO_2 values typically found in the tissues of the body. Here, relatively small changes in PO_2 result in large changes in saturation. This is advantageous because this is the “unloading” portion of the curve where hemoglobin loses its oxygen to the tissues.

Many factors determine the hemoglobin saturation. If, for example, the blood becomes more acidic, the dissociation curve shifts to the right. This indicates that more oxygen is being unloaded from the hemoglobin at the tissue level. This rightward shift of the curve (see [figure 7.11a](#)), attributable to a decline in pH, is referred to as the Bohr effect. The pH in the lungs is generally high, so hemoglobin passing through the lungs has a strong affinity for oxygen, encouraging high saturation. At the tissue level, especially during exercise, the pH is lower, causing oxygen to dissociate from hemoglobin, thereby supplying oxygen to the tissues. With exercise, the ability to unload oxygen to the muscles increases as the muscle pH decreases.

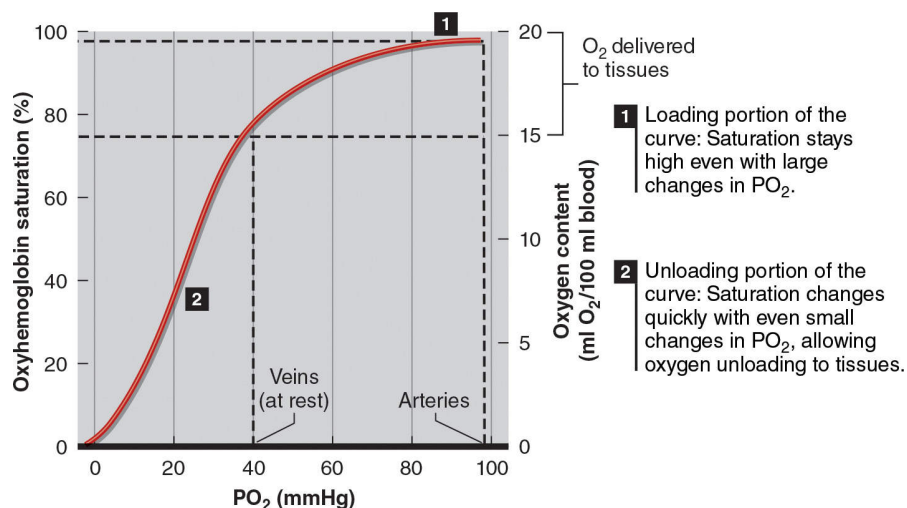


FIGURE 7.10 Oxyhemoglobin dissociation curve.

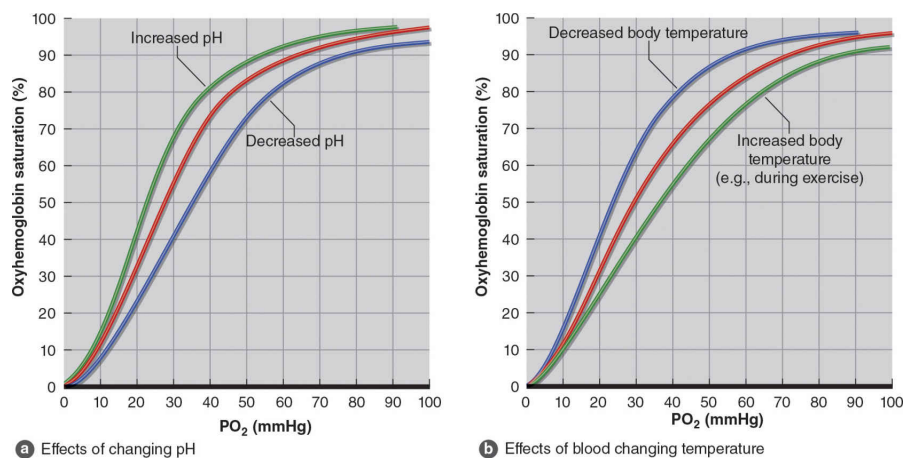


FIGURE 7.11 The effects of (a) changing blood pH and (b) blood temperature on the oxyhemoglobin dissociation curve.

Blood temperature also affects oxygen dissociation. As shown in [figure 7.11b](#), increased blood temperature shifts the dissociation curve to the right, indicating that oxygen is unloaded from hemoglobin more readily at higher temperatures. Because of this, the hemoglobin unloads more oxygen when blood circulates through the metabolically heated active muscles.

Blood Oxygen-Carrying Capacity

The oxygen-carrying capacity of blood is the maximal amount of oxygen the blood can transport. It depends primarily on the blood hemoglobin content. Each 100 ml of blood contains an average of 14 to 18 g of hemoglobin in men and 12 to 16 g in women. Each gram of hemoglobin can combine with about 1.34 ml of oxygen, so the oxygen-carrying capacity of blood is approximately 16 to 24 ml per

100 ml of blood when blood is fully saturated with oxygen. At rest, as the blood passes through the lungs, it is in contact with the alveolar air for approximately 0.75 s. This is sufficient time for hemoglobin to become 98% to 99% saturated. At high intensities of exercise, the contact time is greatly reduced, which can reduce the binding of hemoglobin to oxygen and slightly decrease the saturation, although the unique “S” shape of the curve guards against large drops.

People with low hemoglobin concentrations, such as those with anemia, have reduced oxygen-carrying capacities. Depending on the severity of the condition, these people might feel few effects of anemia while they are at rest because their cardiovascular system can compensate for reduced blood oxygen content by increasing cardiac output. However, during activities in which oxygen delivery can become a limitation, such as highly intense aerobic effort, reduced blood oxygen content limits performance.

Carbon Dioxide Transport

Carbon dioxide also relies on the blood for transportation. Once carbon dioxide is released from the cells, it is carried in the blood primarily in three forms:

- As bicarbonate ions resulting from the dissociation of carbonic acid
- Dissolved in plasma
- Bound to hemoglobin (called carbaminohemoglobin)

Bicarbonate Ion

The majority of carbon dioxide is carried in the form of bicarbonate ion. Bicarbonate accounts for the transport of 60% to 70% of the carbon dioxide in the blood. Carbon dioxide and water molecules combine to form carbonic acid (H_2CO_3). This reaction is catalyzed by the enzyme carbonic anhydrase, which is found in red blood cells. Carbonic acid is unstable and quickly dissociates, freeing a hydrogen ion (H^+) and forming a bicarbonate ion (HCO_3^-):



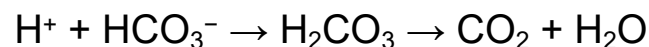
The H^+ subsequently binds to hemoglobin, and this binding triggers the Bohr effect, mentioned previously, which shifts the oxygen–hemoglobin dissociation curve to the right. The bicarbonate ion

diffuses out of the red blood cell and into the plasma. In order to prevent electrical imbalance from the shift of the negatively charged bicarbonate ion into the plasma, a chloride ion diffuses from the plasma into the red blood cell. This is called the chloride shift.

Additionally, the formation of hydrogen ions through this reaction enhances oxygen unloading at the level of the tissue. Through this mechanism, hemoglobin acts as a buffer, binding and neutralizing the H^+ and thus preventing any significant acidification of the blood. Acid–base balance is discussed in more detail in [chapter 8](#).



When the blood enters the lungs, where the PCO_2 is lower, the H^+ and bicarbonate ions rejoin to form carbonic acid, which then dissociates into carbon dioxide and water:



The carbon dioxide that is thus re-formed can enter the alveoli and be exhaled.

Dissolved Carbon Dioxide

Part of the carbon dioxide released from the tissues is dissolved in plasma, but only a small amount, typically just 7% to 10%, is transported this way. This dissolved carbon dioxide comes out of solution where the PCO_2 is low, as in the lungs. There it diffuses from the pulmonary capillaries into the alveoli to be exhaled.

Carbaminohemoglobin

Carbon dioxide transport also can occur when the gas binds with hemoglobin, forming carbaminohemoglobin. The compound is so named because carbon dioxide binds with amino acids in the globin part of the hemoglobin molecule, rather than with the heme group as oxygen does. Because carbon dioxide binding occurs on a different part of the hemoglobin molecule than does oxygen binding, the two processes do not compete. However, carbon dioxide binding varies with the oxygenation of the hemoglobin (deoxyhemoglobin binds carbon dioxide more easily than oxyhemoglobin) and the partial pressure of CO_2 . Carbon dioxide is released from hemoglobin when PCO_2 is low, as it is in the lungs. Thus, carbon dioxide is readily released from the hemoglobin in the lungs, allowing it to enter the alveoli to be exhaled.

In Review

- Oxygen is transported in the blood primarily bound to hemoglobin (as oxyhemoglobin), although a small part of it is dissolved in plasma.
- To better respond to increased oxygen demand, hemoglobin unloading of oxygen (desaturation) is enhanced (i.e., the curve shifts to the right) when
 - PO_2 decreases,
 - pH decreases, or
 - temperature increases.
- Because of the sigmoid shape of the curve, loading of hemoglobin with oxygen in the lungs is only minimally affected by the shift.
- In the arteries, hemoglobin is usually about 98% saturated with oxygen. This is a higher oxygen content than our bodies require, so the blood's oxygen-carrying capacity seldom limits performance in healthy individuals.
- Carbon dioxide is transported in the blood primarily as bicarbonate ion. This prevents the formation of carbonic acid, which can cause H^+ to accumulate and lower the pH. Smaller amounts of carbon dioxide are either dissolved in the plasma or bound to hemoglobin.

Gas Exchange at the Muscles

We have considered how the respiratory and cardiovascular systems bring air into our lungs, exchange oxygen and carbon dioxide in the alveoli, and transport oxygen to the muscles and carbon dioxide to

the lungs. We now consider the delivery of oxygen from the capillary blood to the muscle tissue.

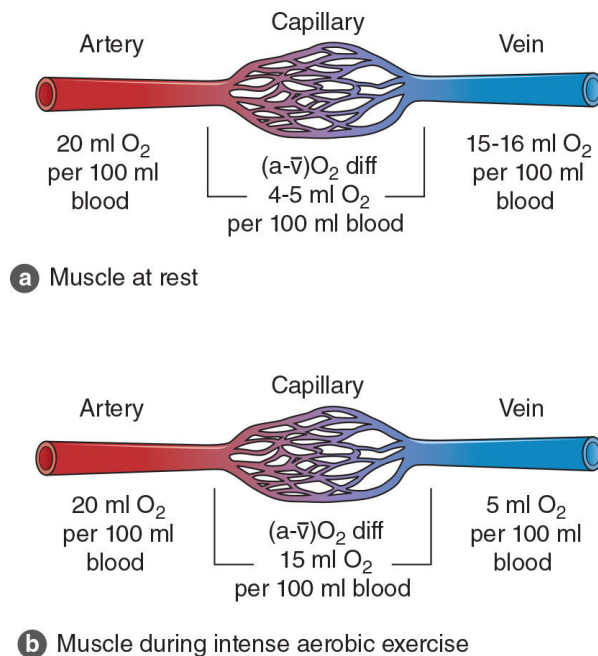
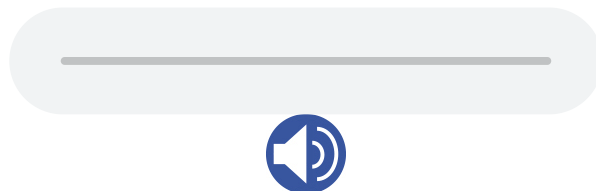


FIGURE 7.12 The arterial–mixed venous oxygen difference, or $(a-\bar{v})O_2$ difference, across the muscle (a) at rest and (b) during intense aerobic exercise.



Arterial–Venous Oxygen Difference

At rest, the oxygen content of arterial blood is about 20 ml of oxygen per 100 ml of blood. As shown in [figure 7.12a](#), this value decreases to 15 to 16 ml of oxygen per 100 ml after the blood has passed through the capillaries into the venous system. This difference in oxygen content between arterial and venous blood is referred to as the **arterial–mixed venous oxygen difference, or $(a-\bar{v})O_2$ difference**. The term *mixed venous* ($\bar{v}$) refers to the oxygen content of blood in the right atrium, which comes from all parts of the body, both active and inactive. The difference between arterial and mixed venous oxygen content reflects the 4 to 5 ml of oxygen per 100 ml of blood taken up by the tissues. The amount of oxygen taken up is

proportional to its use for oxidative energy production. Thus, as the rate of oxygen use increases, the $(a-\bar{v})O_2$ difference also increases. It can increase to 15 to 16 ml per 100 ml of blood during maximal levels of endurance exercise (figure 7.12b). However, at the level of the contracting muscle, the **arterial– venous oxygen difference, or $(a-v)O_2$ difference**, during intense exercise can increase to 17 to 18 ml per 100 ml of blood. Note that there is not a bar over the v in this instance because we are now looking at local muscle venous blood, not mixed venous blood in the right atrium. During intense exercise, more oxygen is unloaded to the active muscles because the PO_2 in the muscles is substantially lower than in arterial blood.

Oxygen Transport in the Muscle

Before oxygen can be used in oxidative metabolism, it must be transported in the muscle to the mitochondria by a molecule called **myoglobin**. Myoglobin is similar in structure to hemoglobin, but myoglobin has a much greater affinity for oxygen than hemoglobin. This concept is illustrated in figure 7.13. At PO_2 values less than 20 mmHg, the myoglobin dissociation curve is much steeper than the dissociation curve for hemoglobin. Myoglobin releases its oxygen content only under conditions in which the PO_2 is very low. Note from figure 7.13 that at a PO_2 at which venous blood is unloading oxygen, myoglobin is loading oxygen. It is estimated that the PO_2 in the mitochondria of an exercising muscle may be as low as 1 mmHg; thus, myoglobin readily delivers oxygen to the mitochondria.

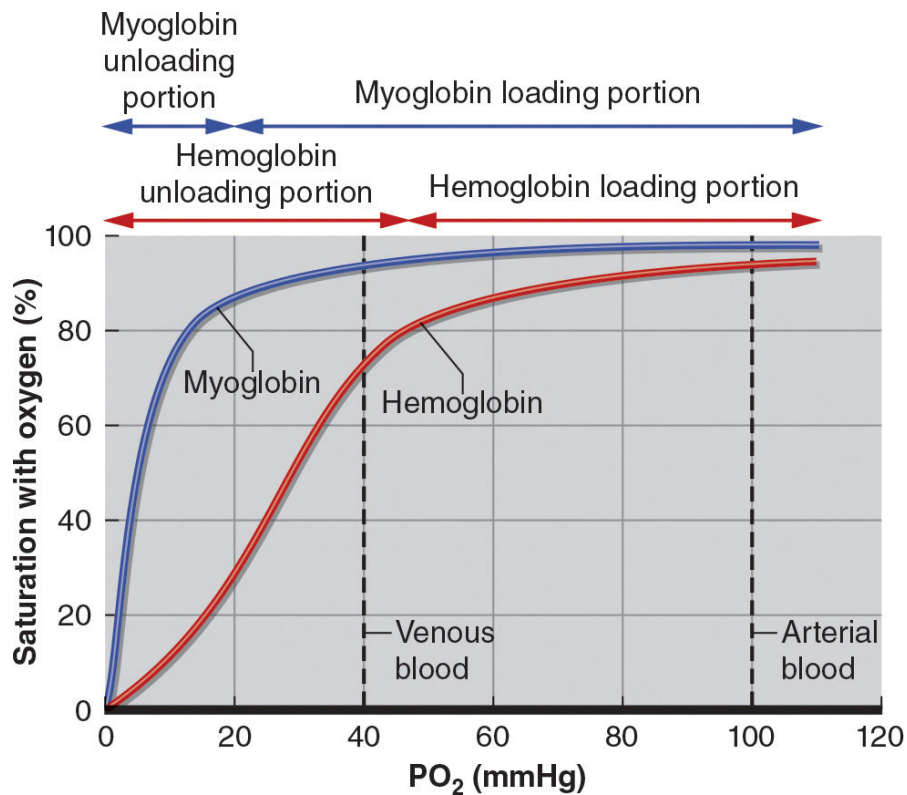


FIGURE 7.13 A comparison of the dissociation curves for myoglobin and hemoglobin.

Factors Influencing Oxygen Delivery and Uptake

The rates of oxygen delivery and uptake depend on three major variables:

- Oxygen content of blood
- Blood flow
- Local conditions (e.g., pH, temperature)

With exercise, each of these variables is adjusted to ensure increased oxygen delivery to active muscle. Under normal circumstances, hemoglobin is about 98% saturated with oxygen. Any reduction in the blood's normal oxygen-carrying capacity would hinder oxygen delivery and reduce cellular uptake of oxygen. Likewise, a reduction in the PO_2 of the arterial blood would lower the partial pressure gradient, limiting the unloading of oxygen at the tissue level. Exercise increases blood flow through the muscles. As more blood carries oxygen through the muscles, less oxygen must be removed from each 100 ml of blood (assuming the demand is unchanged). Thus, increased blood flow improves oxygen delivery.

Many local changes in the muscle during exercise affect oxygen delivery and uptake. For example, muscle activity increases muscle acidity because of lactate production. Also, muscle temperature and carbon dioxide concentration both increase because of increased metabolism. All these changes increase oxygen unloading from the hemoglobin molecule, facilitating oxygen delivery and uptake by the muscles.

Carbon Dioxide Removal

Carbon dioxide exits the cells by simple diffusion in response to the partial pressure gradient between the tissue and the capillary blood. For example, muscles generate carbon dioxide through oxidative metabolism, so the PCO_2 in muscles is relatively high compared with that in the capillary blood. Consequently, CO_2 diffuses out of the muscles and into the blood to be transported to the lungs.

In Review

- The $(a-\bar{V})O_2$ difference is the difference in the oxygen content of arterial and mixed venous blood throughout the body. This measure reflects the amount of oxygen taken up by the tissues, active and inactive.
- The $(a-\bar{V})O_2$ difference increases from a resting value of about 4 to 5 ml per 100 ml of blood up to values of 18 ml per 100 ml of blood during intense exercise. This increase reflects an increased extraction of oxygen from arterial blood by active muscle, thus decreasing the oxygen content of the venous blood.
- Oxygen delivery to the tissues depends on the oxygen content of the blood, blood flow to the tissues, and local conditions (e.g., tissue temperature and PO_2).
- Within muscle, oxygen is transported to the mitochondria by a molecule called myoglobin. Compared to the oxyhemoglobin dissociation curve, the myoglobin- O_2 dissociation curve is much steeper at low PO_2 values.
- Myoglobin releases its oxygen only at a very low PO_2 . This is compatible with the PO_2 found in exercising muscle, which may be as low as 1 mmHg.
- Carbon dioxide exchange at the tissues is similar to oxygen exchange, except that carbon dioxide leaves the muscles, where it is formed, and enters the blood to be transported to the lungs for clearance.

Regulation of Pulmonary Ventilation

Maintaining homeostatic balance in blood PO_2 , PCO_2 , and pH requires a high degree of coordination between the respiratory, muscular, and circulatory systems. Much of this coordination is accomplished by involuntary regulation of pulmonary ventilation. This control is not yet fully understood, although many of the intricate neural controls have been identified.

The respiratory muscles are under the direct control of motor neurons, which are in turn regulated by **respiratory centers** (inspiratory and expiratory) located within the brain stem (in the medulla oblongata and pons). These centers establish the rate and depth of breathing by sending out periodic impulses to the respiratory muscles. The cortex can override these centers if voluntary control of respiration is desired. Additionally, input from other parts of the brain occurs under certain conditions.

The inspiratory area of the brain (dorsal respiratory group) contains cells that intrinsically fire and control the basic rhythm of ventilation. The expiratory area is quiet during normal breathing (recall that expiration is a passive process at rest). However, during forceful breathing such as during exercise, the expiratory area actively sends signals to the muscles of expiration. Two other brain centers aid in the control of respiration. The apneustic area has an excitatory effect on the inspiratory center, resulting in prolonged firing of the inspiratory neurons. Finally, the pneumotaxic center inhibits or switches off inspiration, helping to regulate inspiratory volume.

RESEARCH PERSPECTIVE 7.3

Ventilation During Exercise in Asthma

Asthma, a condition in which the airways are inflamed and narrowed, changes airway function and makes breathing difficult. Because these changes in airway function are variable, asthmatics experience daily fluctuations in airway inflammation, airway hyper-responsiveness, pulmonary function, and clinical symptoms. Regular aerobic exercise is recommended for asthmatics, and asthmatics who are physically active show improvements in exercise capacity. However, despite a large body of literature characterizing exercise-induced bronchoconstriction in asthmatics, a significant gap in knowledge exists with regard to the influences of variable airway function at rest on the responses to aerobic exercise.

A recent study sought to determine the effects of both improved and worsened preexercise airway mechanical function on the ventilatory responses to aerobic exercise in asthmatic and nonasthmatic adults.⁵ All subjects completed four separate exercise bouts of 3 min of cycling at 70% of their peak workload, followed by continuous exercise at 85% of peak workload until volitional exhaustion. Each exercise bout was preceded by one of four different interventions: (1) inhalation of a fast-acting β_2 -agonist to improve airway function, (2) a eucapnic voluntary hyperpnea challenge to worsen airway function, (3) a sham version of the hyperpnea, and (4) a control trial. Pulmonary function was assessed using an automated spirometer.

Surprisingly, despite markedly different preexercise pulmonary function (experimentally manipulated by each intervention) in asthmatic adults, exercise ventilation was nearly identical among the four conditions. Moreover, there were no differences in exercise ventilation between asthmatic and nonasthmatic adults during any of the four different interventions. These data demonstrate that the pulmonary system of asthmatic adults is capable of adequately responding to the acute demand for increased airflow necessitated by high-intensity aerobic exercise. Clinically, the findings of this study support the notion that habitual aerobic exercise is beneficial for adults with asthma.

The respiratory centers do not act alone in controlling breathing. Breathing also is regulated and modified by the changing chemical environment in the body. For example, sensitive areas in the brain respond to changes in carbon dioxide and H^+ levels. The central chemoreceptors in the brain are stimulated by an increase in H^+ ions in the cerebrospinal fluid. The blood–brain barrier is relatively impermeable to H^+ ions or bicarbonate. However, CO_2 readily diffuses across the blood–brain barrier and then reacts to increase H^+ ions. This, in turn, stimulates the inspiratory center, which then activates the neural circuitry to increase the rate and depth of respiration. This increase in respiration, in turn, increases the removal of carbon dioxide and H^+ .

Chemoreceptors in the aortic arch (the aortic bodies) and in the bifurcation of the common carotid artery (the carotid bodies) not only are sensitive primarily to blood changes in PO_2 but also respond to changes in H^+ concentration and PCO_2 . The carotid chemoreceptors are more sensitive to changes in H^+ concentrations and PCO_2 . Overall, PCO_2 appears to be the strongest stimulus for the regulation of breathing. When carbon dioxide levels become too high, carbonic acid forms, then quickly dissociates, giving off H^+ . If H^+ accumulates,

the blood becomes too acidic (pH decreases). Thus, an increased PCO_2 stimulates the inspiratory center to increase respiration—not to bring in more oxygen but to rid the body of excess carbon dioxide and limit further pH changes.

In addition to the chemoreceptors, other neural mechanisms influence breathing. The pleurae, bronchioles, and alveoli in the lungs contain stretch receptors. When these areas are excessively stretched, that information is relayed to the expiratory center. The expiratory center responds by shortening the duration of an inspiration, which decreases the risk of overinflating the respiratory structures. This response is known as the Hering-Breuer reflex.

Many control mechanisms are involved in the regulation of breathing, as shown in [figure 7.14](#). Such simple stimuli as emotional distress or an abrupt change in the temperature of the surroundings can affect breathing. But all these control mechanisms are essential. The goal of respiration is to maintain appropriate levels of the blood and tissue gases as well as proper pH for normal cellular function. Small changes in any of these, if not carefully controlled, could impair physical activity and jeopardize health.



Afferent Feedback From Exercising Limbs

The respiratory system responds almost immediately to increased ventilation at the initiation of exercise, even before there is a significant increase in the metabolic demand from exercising muscle.

The fast initiation of the drive to breathe results from a combination of central command (the brain's feedforward mechanism) and afferent neural feedback from the working limbs.

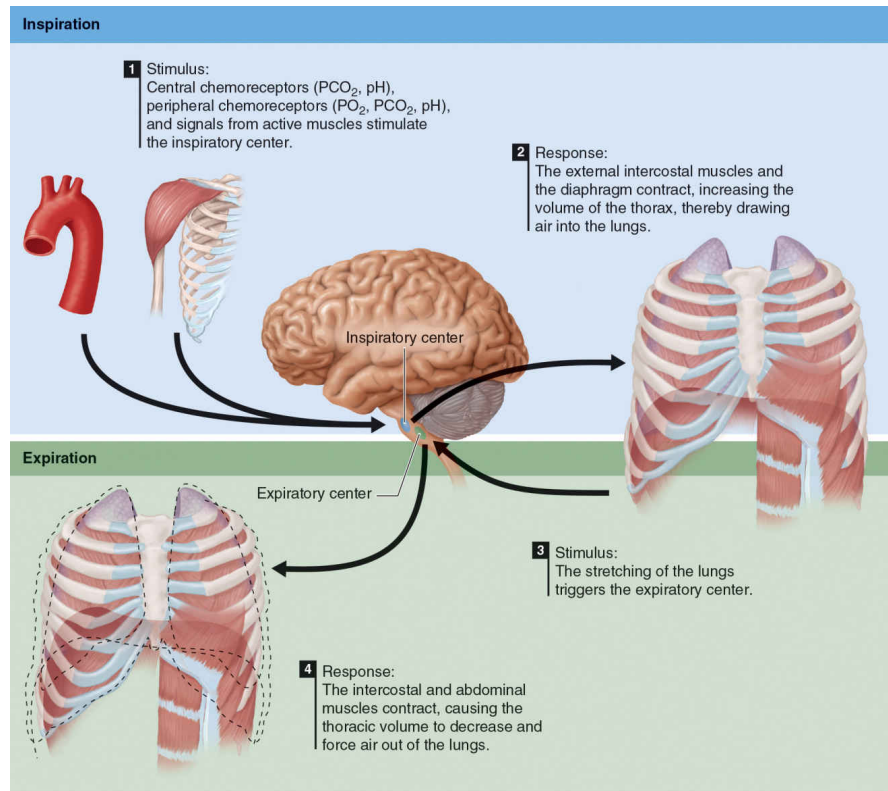
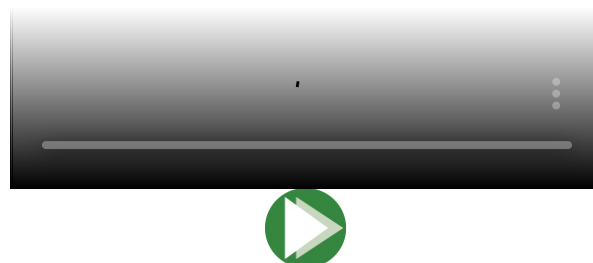


FIGURE 7.14 An overview of the processes involved in respiratory regulation.



In addition to those physiological mechanisms, it has been shown that the fast drive to breathe at the beginning of exercise is proportional to the frequency of limb movement. In attempting to separate the contributions to the control of ventilation from central

command and afferent feedback from locomotor muscles, ventilation was measured in a group of subjects as they ran at two different speeds on a treadmill.¹ When the subjects started running at a given constant speed, their ventilation immediately increased in proportion to the treadmill speed. However, when subjects began running at a lower speed, but with the grade elevated to match the workload of the faster flat (0 grade) condition, their ventilation first increased to match the slower speed and then gradually drifted up to meet their actual oxygen demand. The immediate increase in ventilation was partially controlled by afferent feedback from the limbs, but the subsequent gradual increase in ventilation suggested that increased ventilation is a response to metabolic changes and increased metabolic demand from the exercising muscle.

RESEARCH PERSPECTIVE 7.4

Regular Exercise Reduces Respiratory Disease Mortality

Pneumonia, an infection that causes inflammation of the air sacs in the lungs, is the leading cause of infection-related death in the United States. The risk of pneumonia increases with age and comorbid conditions such as heart disease, chronic lung disease, and use of immunosuppressive drugs. Multiple health benefits have been attributed to regular physical activity; however, it remains unclear whether these benefits extend to decreased risk of respiratory disease. Certainly, reductions in the risk of pneumonia would be consistent with concept that regular exercise prevents age-related declines in function.

A 2014 report examined the association of running and walking with mortality due to respiratory disease in the National Walkers' and Runners' Health Studies, a prospective epidemiological cohort of over 150,000 adults.⁶ This large cohort was used to test the hypothesis that greater exercise energy expenditure would be associated with a lower risk for respiratory diseases in general and pneumonia in particular. The results provided strong support for a reduction in the risk for respiratory diseases and pneumonia as underlying and contributing causes of mortality with greater exercise energy expenditure. Not surprisingly, this relation was dose dependent, with more substantial reductions in risk occurring in those with greater levels of habitual activity. Interestingly, this risk reduction was not different between walkers and runners. In addition, these effects appear to be independent of the effects of exercise on cardiovascular disease risk. These findings add to the compelling evidence for the health benefits of regular aerobic exercise.

More recently, scientists have been interested in whether afferent neural feedback from the limbs continues throughout exercise. Investigators at the University of Toronto had subjects independently alter either their pedal cadence or resistance while cycling during two different trials.² During one, they varied their pedal speed in a sinusoidal manner while keeping their total workload constant, and during the other one, they kept their speed constant while varying their pedal workload sinusoidally (see [figure 7.15](#)). During the trial in which pedal speed varied ([figure 7.15a](#)), there was a much faster increase in ventilation that preceded any changes in heart rate. In contrast, when subjects altered their workload ([figure 7.15b](#)) but kept their pedal speed constant, there was a greater lag time before the increase in ventilation, such that the metabolic changes preceded changes in ventilation. The results from these unique experiments suggest that limb movement frequency influences ventilation at the start of, and throughout, exercise. Continued afferent neural feedback from the limbs influences the drive to breathe during exercise.

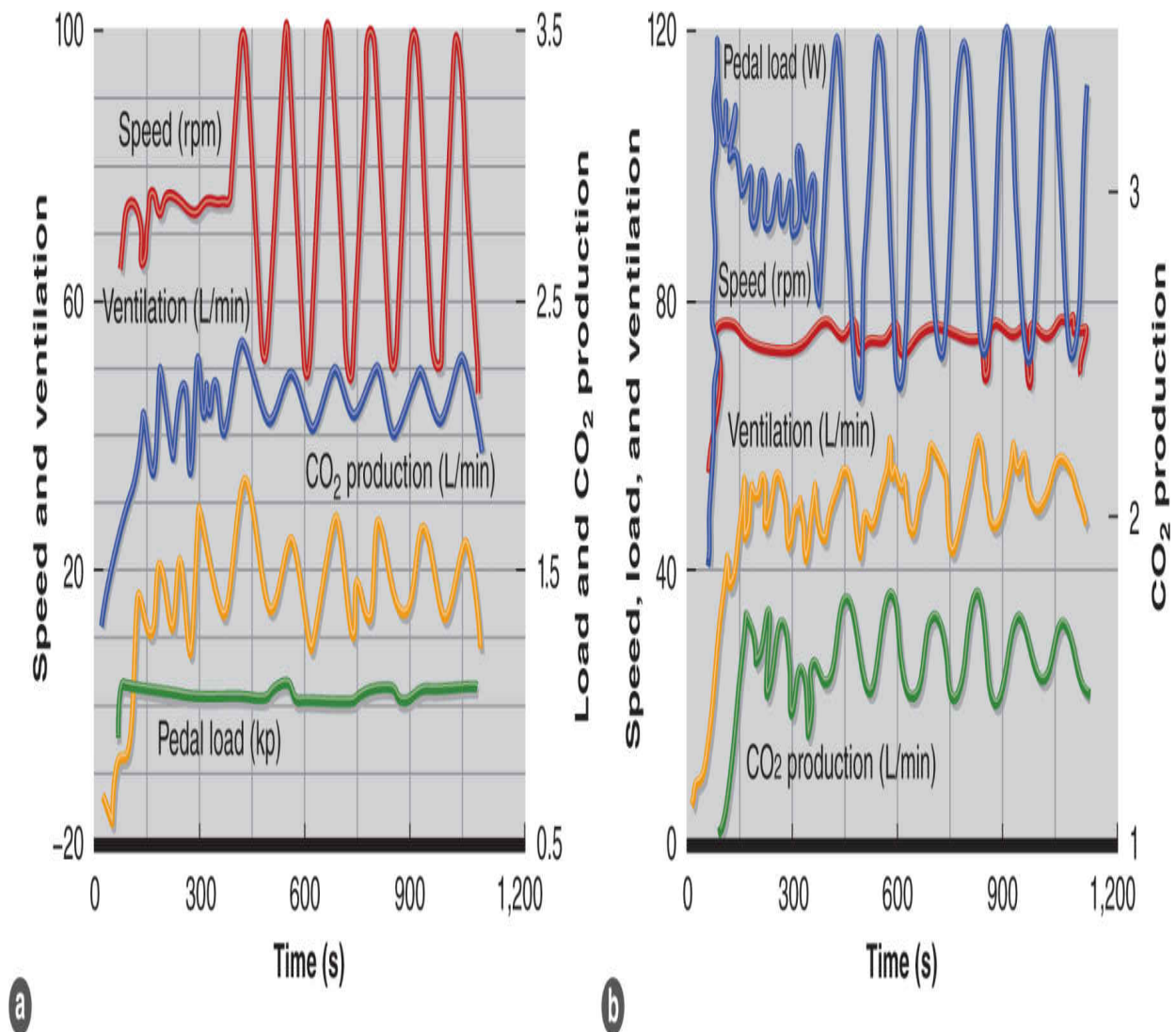


FIGURE 7.15 Sine wave exercise experiments. (a) Breath-by-breath variables measured during an exercise test with the subject varying pedaling speed (cadence) while pedal loading remains constant. The solid lines are fitted sine waves. (b) Breath-by-breath variables measured during an exercise test with varying pedal loading while pedaling speed (cadence) remains constant.

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IN CLOSING

In [chapter 6](#), we discussed the role of the cardiovascular system during exercise. In this chapter, we looked at the role played by the respiratory system. The entire process of respiration involves pulmonary ventilation (inspiration and expiration), diffusion of gases at the alveoli, transport of gases through the blood, and gas exchange at the tissues. In the next chapter, we examine how the cardiovascular and respiratory systems respond to an acute bout of exercise.

KEY TERMS

alveoli

arterial–mixed venous oxygen difference, or $(a-\bar{V})O_2$ difference

arterial–venous oxygen difference, or $(a-v)O_2$ difference

Boyle's gas law

Dalton's law

dead space

expiration

external respiration

Fick's law

Henry's law

inspiration

internal respiration

myoglobin

oxygen cascade

oxygen diffusion capacity

partial pressure

pulmonary diffusion

pulmonary ventilation

residual volume (RV)

respiratory centers

respiratory membrane

respiratory pump

spirometry

tidal volume

total lung capacity (TLC)

vital capacity (VC)

STUDY QUESTIONS

1. Describe and differentiate between external and internal respiration.
2. Describe the mechanisms involved in inspiration and expiration.
3. What is a spirometer? Describe and define the lung volumes measured using spirometry.
4. Explain the concept of partial pressures of respiratory gases—oxygen, carbon dioxide, and nitrogen. What is the role of gas partial pressures in pulmonary diffusion?
5. Where in the lung does the exchange of gases with the blood occur? Describe the role of the respiratory membrane.
6. How are oxygen and carbon dioxide transported in the blood?
7. Describe the oxygen cascade from dry ambient air to the tissues and into the venous circulation. Provide appropriate values for the various partial pressures of oxygen at each level.
8. How is oxygen unloaded from the arterial blood to the muscle and carbon dioxide removed from the muscle into the venous blood?
9. What is meant by the arterial–mixed venous oxygen difference? How and why does this change from resting conditions to exercise conditions?
10. Describe how pulmonary ventilation is regulated. What are the chemical stimuli that control the depth and rate of breathing? How do they control respiration during exercise?

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.

